



CSE471: System Analysis and Design
Assignment on Functional Requirements
Proposed Project Title: *Decorate3D: Smart C2C Used-Furniture & Interior Design Marketplace* [Sample]

Group No: 04, CSE471 Lab Section: 14, Summer 2025	
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Tech Stack:

- Language: Typescript
- Framework: React.js (Frontend library) & Node.js with Express.js (Backend server framework)
- Styling: Tailwind CSS & DaisyUI
- Database: MongoDB
- ODM: Mongoose
- Deployment: Vercel

Functional requirements:

Module 1

1. [Md.Ashfaq Habib Rafi] - Sellers can upload multiple high-resolution photos of their used furniture to obtain an AI-generated damage assessment. The system integrates with an external AI vision API (such as OpenAI GPT-4o Vision) to automatically scan the images for visible tears, scratches, dents, or structural damage and generates an objective condition grade (Fair, Good, Excellent) alongside an annotated image overlay marking the damage locations.
2. [Muhtasim Ahmed] - Buyers can view and interact with 3D models of listed furniture items. The system renders an interactive, 360-degree rotatable 3D canvas of the item on the product details page (using a React-based Three.js renderer) so that buyers can inspect structural elements and aesthetic design from all angles before purchasing.
3. [Injamamul Haque Fahim] Buyers can search and locate furniture listings within a local geographical radius. The system integrates the Google Maps JavaScript API to display active seller listings as pins on an interactive map based on the buyer's current GPS location and custom search radius settings.
4. [Shouvik Banik] Buyers can make secure credit card payments during checkout. The system integrates the Stripe API to process the transaction, split and collect the platform's transaction commission, and securely hold the remaining payout in an escrow holding state.

Module 2

1. [Md.Ashfaq Habib Rafi] - Sellers can receive dynamic, data-driven pricing recommendations for their items. The system processes the original purchase price, item age, category, and the AI-generated condition grade to execute a local regression model and suggest a competitive listing price range.
2. [Muhtasim Ahmed] - Users can layout and test furniture placements on a virtual 3D floor plan. The system provides an interactive grid editor where users input room dimensions, draw walls/doors, and drag-and-drop 3D models of listed items to verify spatial fitting and clearance before purchasing.
3. [Injamamul Haque Fahim] Buyers can check live shipping price quotes before purchasing. The system integrates the Google Distance Matrix API to calculate the exact distance and driving duration between the seller and buyer, generating a dynamic courier quote based on travel distance and item dimensions.
4. [Shouvik Banik] - Users can secure their funds in a safe escrow system. The system automatically locks the buyer's payment in an escrow state upon checkout and prevents any payout release to the seller or courier until physical delivery is successfully completed.

Module 3

1. [Md.Ashfaq Habib Rafi] - Buyers can perform visual style searches to locate similar furniture items. The system allows users to upload inspiration images and uses visual semantic vector search in MongoDB Atlas to retrieve active listings that match the style, color, and structure.
2. [Md.Ashfaq Habib Rafi] - Sellers can list items faster using automatic attribute tagging. The system processes listing photos to classify the item and auto-fill tags such as item category (sofa, chair, table), material, and color during listing creation.
3. [Muhtasim Ahmed] - Buyers can project 3D furniture models into their physical rooms. The system integrates the WebXR Device API and Three.js, prompting the user for camera access and anchoring a 1:1 scale model of the furniture in the live camera feed.
4. [Muhtasim Ahmed] - Buyers can verify physical clearance spaces using automated camera tools. The system uses WebXR camera depth sensing to measure doorway and room widths, comparing them with the furniture dimensions and overlays color-coded fit alerts (Green for fits, Red for collision).

5. [Injamamul Haque Fahim] - Independent couriers can search and accept local transport jobs. The system provides an interactive dispatch board listing the pickup/drop-off coordinates and item dimensions of sold furniture, allowing verified drivers to bid on or lock jobs.
6. [Injamamul Haque Fahim] - Buyers and sellers can track the real-time location of the delivery courier. The system integrates the Google Directions API to render a live tracking route map that displays the courier's GPS location and estimated time of arrival during transit.
7. [Shouvik Banik] - Buyers and couriers can secure delivery handovers using OTP and QR verification. When the courier completes delivery, they scan the buyer's dynamic QR code or input the OTP to unlock the transaction, which automatically generates a detailed PDF invoice and transaction receipt.
8. [Shouvik Banik] - Users can open formal disputes if delivered items do not match listing descriptions. The system locks the escrow funds in the dispute state and provides an interactive portal for buyers and sellers to upload photographic proof and chat, with escalation options for admin mediation.